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Vertical and lateral preferential flow in hillslope soils of *Cryptomeria japonica* plantations within strip-cutting areas

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A R T I C L E I N F O

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A B S T R A C T

Vertical and lateral preferential flow are key regulators of hillslope hydrology, but how harvesting disturbance alters these flow pathways in relation to soil properties and root traits remains uncertain. In a *Cryptomeria japonica* plantation (23-years-old) in Sichuan, China, soil samples were collected from 0 to 50 cm depths at upslope, midslope, and downslope positions in various strip-cutting (30 m cutting + 45 m retained, 40%, intensity) microsites (MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area) and in an uncut control area (UC). Dye-tracing experiments were conducted to quantify vertical and lateral preferential flow, and to examine their relations with soil properties and root traits. The results indicated that soil properties and root traits varied with soil depth and site type. Soil organic matter content varied among sites across slope positions and soil layers, whereas initial soil water content was lower in EC than in UC across slope positions. Live root abundance (LRA) was greater in UC than in MC in 0–10 cm across all slope positions, whereas dead root abundance (DRA) in MC and EC exceeded that in UC in the 0–30 cm at the downslope position. The dye coverage of vertical preferential flow (VDC) was lower in MC than in UC across most soil layers at upslope and midslope, whereas the maximum downslope transport distance of lateral preferential flow (LD_{max}) in MC exceeded that in UC at the downslope. Among soil properties and root traits, LRA had a significant positive effect on VDC but was negatively correlated with LD_{max}, whereas DRA showed the opposite pattern. These findings highlight the importance of changes in root status induced by strip-cutting in regulating vertical and lateral preferential flow pathways, and indicate that downslope MC represents a key area for managing hydrological risks.

1. Introduction

Preferential flow, which occurs through macropores that occupy only a small fraction of the total soil volume, enables water and solutes to reach greater depths more rapidly (Allaire et al., 2009; Gerke et al., 2015; Wu et al., 2021; Luo et al., 2023). This type of flow, which does not strictly follow Darcy's law, rapidly moves rapidly along preferential pathways (Beven and Germann, 1982). On hillslopes, preferential flow can notably shape subsurface drainage, influence streamflow patterns, alter soil erosion and slope stability, affect how hillslopes respond to precipitation, and control the transport of nutrients, pesticides, and pathogens (Shao et al., 2015; Wiekenkamp et al., 2016; Zhang et al., 2024). As critical units for water conservation and hydrological regulation, forest ecosystems, particularly those subjected to harvesting disturbances, have received sustained attention in recent years because of their effects on hillslope hydrological processes (Zhao et al., 2021; Tigabu et al., 2025). Therefore, identifying and quantifying preferential flow on harvesting-disturbed hillslopes is vital to understanding hillslope hydrological processes and their implications for forest health (Beven and Germann, 2013; Mei et al., 2018; Wen et al., 2025).

As integral components of subsurface flow, vertical and lateral preferential flow, occur simultaneously and move through soil macropore networks on hillslopes, thereby substantially influencing surface-groundwater exchange (Anderson et al., 2009a, 2009b; McGuire et al., 2024). Lateral preferential flow through a self-organizing network of macro- and mesopores in the soil matrix is critical for connecting soil

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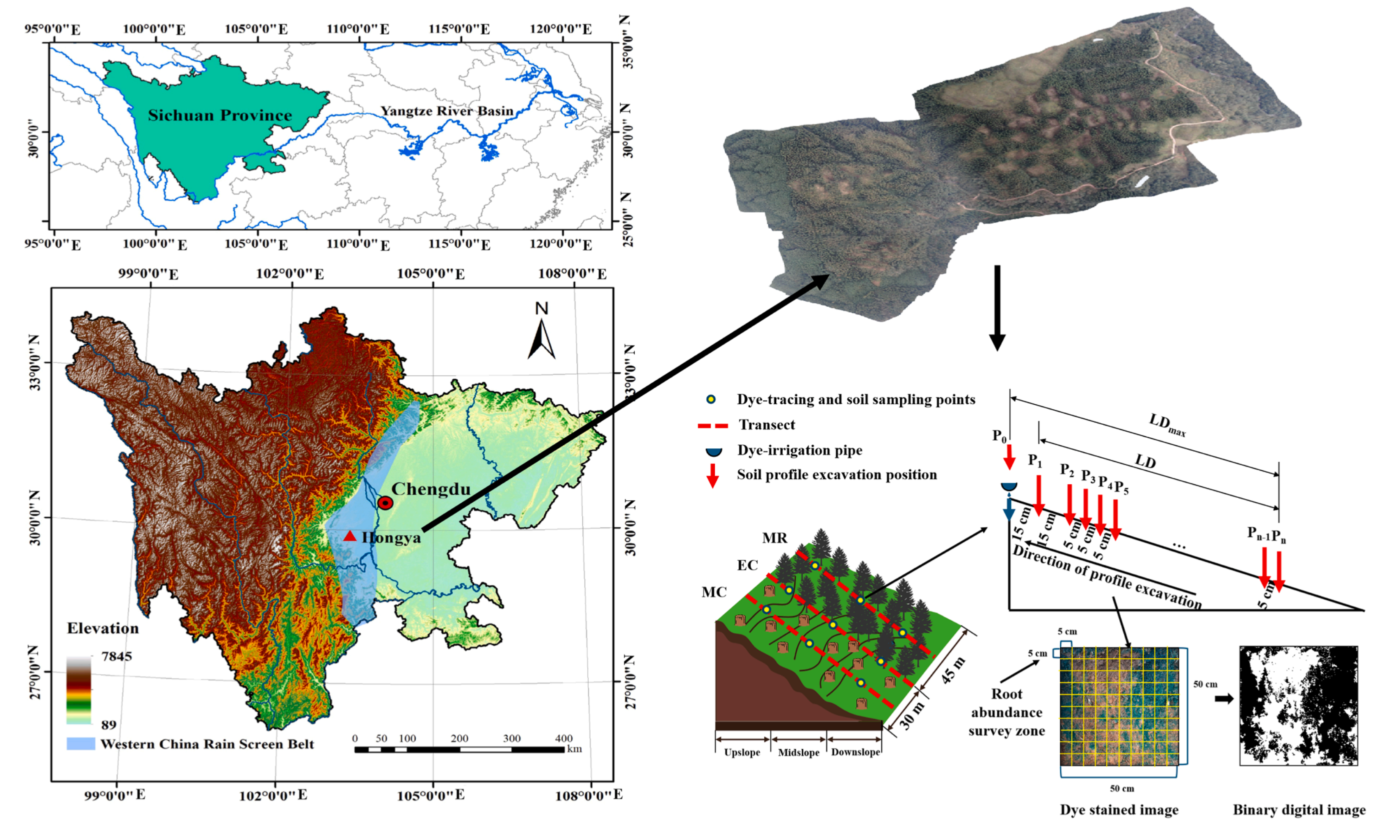

Fig. 1. Geographical location of research area and experimental layout. MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; Where P_0–P_n refer to the dye-stained soil profiles (P) excavated along the downslope direction, with P_0 representing the profile directly beneath the irrigation pipe and P_n representing the farthest dye-stained profile from the pipe. P_0 was used to analyse vertical preferential flow, whereas P_1–P_n were used to analyse lateral preferential flow. LD represents the cumulative downslope distance from P_1 to P_n used for calculating lateral flow volume density. LD_{max} was defined as the downslope distance between P_0 and P_n, representing the maximum downslope transport distance of lateral preferential flow.

development, hillslope hydrology, internal erosion, slope stability, and biogeochemical export processes in forested mountain ecosystems (Buttle and McDonald, 2002; Leslie and Heinse, 2013; Bourgault et al., 2017; Nespoulous et al., 2019; Y. Xu et al., 2025). Recently, electrical resistivity tomography (ERT) (Leslie and Heinse, 2013; Jiang et al., 2026), soil moisture monitoring (Kaffas et al., 2025), the hydrometric tracer approach (Buttle and McDonald, 2002; van der Heijden et al., 2013), and indoor soil tanks have been applied to study vertical and lateral flow (Y. Xu et al., 2025). Numerous studies have demonstrated that vertical preferential flow is controlled by the vegetation type, soil properties, soil faunal activity, geographic setting, and root traits (e.g., orientation, length, amount, diameter, status and architectural complexity of roots) (Bi et al., 2025; Cai et al., 2024; Wang et al., 2024). Furthermore, Lu et al. (2024) reported that the degree of lateral flow was greater under wet initial moisture conditions than under dry initial moisture conditions. Y. Xu et al. (2025) observed that finer roots enhanced lateral flow transport, whereas coarser roots more significantly facilitated vertical flow. Nespoulous et al. (2019) indicated that the effects of fine roots on vertical and lateral flow are more notable than those of coarse roots and macrofaunal activity. Compared with other methods, in situ Brilliant Blue dye-tracing experiments are more cost-effective and environmentally friendly, producing bright, visually distinct pathways that clearly reveal both types of preferential flow patterns (Anderson et al., 2009b; Gerke et al., 2015; Nespoulous et al., 2019). Nevertheless, studies using this method to simultaneously investigate vertical and lateral preferential flow remain limited. Furthermore, few investigations have simultaneously addressed how soil properties and root traits influence these flow paths on forested hillslopes.

Strip-cutting is defined as a harvesting method in which strips of uncut forests are retained within the harvested area. Compared with large-scale clear-cutting, strip cutting can retain part of the operational efficiency of timber harvesting while reducing ecological disturbance and maintaining important ecosystem functions (Baker et al., 2015; Fedrowitz et al., 2014; Ito et al., 2006; Lv et al., 2024; Veysey Powell and Babbitt, 2017). Strip clear-cutting creates hillslope-scale microsites comprising harvested strips, edge transition zones, and retained strips, thus generating distinct spatial heterogeneity in both microenvironmental conditions and surface disturbance (Plaugher and Schuler, 2025; Song et al., 2025). Unlike clear cutting, strip-cutting has been demonstrated to maintain a greater share of well-drained areas (Stenberg et al., 2022). Furthermore, timber harvesting consistently decreases the soil saturated hydraulic conductivity and increases the bulk density (Behringer et al., 2025a, 2025b). Wu et al. (2021) also reported that after thinning, decayed roots maintained a high, stable infiltration rate, thereby effectively increasing soil water replenishment in dry forest soil layers. However, in harvesting areas, especially those managed under strip-cutting regimes, vertical and lateral preferential flow and their controlling factors remain poorly understood.

Cryptomeria japonica (*C. japonica*) serves as the dominant afforestation species in the rainy areas of western China, which are located in the upper part of the Yangtze River basin. Owing to poor tree growth, sparse understory vegetation, and decreasing biodiversity, plantations have been managed via the strip-cutting approach at an intensity of 40% (Song et al., 2025). On the basis of the layout characteristics of strip-cutting management, we aimed to answer the following two questions. First, do vertical and lateral preferential flow patterns differ among microsites created by strip-cutting? Second, how do soil properties and root traits influence these flow patterns? This study aimed to advance the understanding of vertical and lateral preferential flow, and

Table 1
Basic soil physical and chemical properties of study sites. (AV ± SE, number=45).

site	Sand (%) (2—0.02 mm)	Silt (%) (0.02—0.002 mm)	Clay (%) (<0.002 mm)	Non-capillary porosity (%)	Total capillary porosity (%)	Total Nitrogen (g/kg)	Total phosphorus (g/kg)
MC	58.95 ± 3.63 A	23.84 ± 1.87 A	17.21 ± 2.04 A	4.70 ± 0.29 A	48.24 ± 5.67 A	1.15 ± 0.06 A	0.20 ± 3.79E−03A
EC	54.92 ± 4.37 A	25.47 ± 2.15 A	19.61 ± 2.27 A	5.74 ± 0.84 A	56.56 ± 2.64AB	1.47 ± 0.09 A	0.25 ± 1.66E−03A
MR	59.44 ± 1.84 A	24.00 ± 1.86 A	16.65 ± 0.13 A	5.18 ± 0.64 A	55.12 ± 2.49AB	1.19 ± 0.02 A	0.20 ± 3.14E−03A
UC	48.92 ± 1.34 A	28.76 ± 0.68 A	22.32 ± 0.68 A	6.10 ± 0.45 A	65.84 ± 1.92B	1.22 ± 0.01 A	0.22 ± 3.63E−03A

Note: MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; UC, uncut control area; AV is average, SE is standard error. Different capital letters indicate significant differences among sites ($P < 0.05$).

to provide a hydrological basis for reducing risks to soil, water, and nutrient dynamics during post-harvest forest management.

2. Materials and methods

2.1. Study area

This study was conducted at the Long-Term Artificial Forest Ecosystems Station of Sichuan Agricultural University in Hongya Forest Farm (29°24′–30°00′ N, 102°49′–103°32′ E), Sichuan Province, China. The station, with covers approximately 10.8 hm^2, and the elevation ranges from 1200 to 1300 m above sea level. The station is subject to a subtropical humid climate, with an average annual temperature of 16.6 °C and a mean annual precipitation of 1435.5 mm. The lowest (4.6 °C) and highest (28.9 °C) mean temperatures occur in the winter and summer, respectively. Soils in the study area are mountain yellow soil, with a pH of approximately 4.5. The main plantation species are *Cryptomeria japonica* var. Sinensis and *Cunninghamia* lanceolata (Lamb.) Hook. The understory vegetation is mainly composed of *Rubus buergeri*, *Vaccinium mandarinorum*, *Dryopteris fuscipes*, *Elatostema involucratum, and Pellionia radicans.*

2.2. Experimental design

In 2020, our research team implemented strip-cutting at an intensity of 40% intensity in 23-year-old *C. japonica* plantations at Hongya Forest Farm, with uncut areas serving as controls. Before strip-cutting, these stands had similar site conditions and stand growth, and the stumps of harvested trees were retained. Subsequently in 2021, the strip-cutting area was replanted with 2- to 3-year-old native broadleaved *Schima superba* at a spacing of 2.5 m. In 2023, we established three replicate study sites respectively in the strip-cutting area (30 m cutting + 45 m retained), as well as in an uncut area (uncut control, UC). The slope gradients in the strip-cutting area ranged from 20° to 23°, whereas those in UC ranged from 21° to 25°. In accordance with the layout features of strip-cutting, three 20 m × 20 m sampling plots with similar slope aspects were established in both cutting area and retained areas, and another three plots of the same size were established in UC. Slope-parallel transects extending from the upper slope to the lower slope were established at the middle of the cutting area (MC), the edge of the cutting area (EC) (The boundary between the cutting area and the retained area), and the middle of the retained area (MR) (Fig. 1). Moreover, three transects were also established at the middle of the three plots in the UC site. These transects were used to support subsequent investigations of preferential flow, soil properties, and root traits in different sites. Three replicate transects were established for each site type, resulting in a total of 12 transects in this study. The basic soil conditions at the study sites are listed in Table 1.

2.3. Dye-tracing experiment setup and soil sample collection

Dye-tracing experiments were conducted, and soil samples were collected at the upslope, midslope, and downslope positions along each transect in four sites (MC, EC, MR, and UC). Therefore, for each slope position within each site type, three dye-tracing points were established to analyse vertical and lateral preferential flow features, and three soil sampling points were established to analyse soil properties and root traits (Fig. 1). Before the dye-tracing experiments were conducted, the litter layer was removed, and herbaceous plants were cut with scissors without disturbing the soil structure. Subsequently, a pipe (60 cm long × 10 cm wide) with 22 holes (2 mm in diameter) was placed parallel to the ground surface. The pipe was positioned approximately 5 cm above the ground surface (Nespoulous et al., 2019). In this study, 30 L of a dye solution (Brilliant Blue, 5 g/L) was prepared for irrigation at a constant discharge rate of 5.0 mL/s (Gerke et al., 2015; Nespoulous et al., 2019). The constant discharge was maintained constant via a pump to ensure that the dye solution seeped through the pipe and that no obvious ponding occurred throughout the dye experiment. Each dye-tracing point was subsequently covered with a plastic sheet to prevent evaporation and rainfall infiltration (Luo et al., 2019). After 5 h of dye irrigation, a soil profile (60 cm wide × 50 cm deep) was excavated beginning at a distance of 100 cm downslope from the irrigation pipe (Gerke et al., 2015; Nespoulous et al., 2019). This distance was determined from preliminary experiments as the maximum reach of the Brilliant Blue tracer in the study area. Excavation of soil profiles then proceeded uphill at 5 cm intervals until reaching the 30 cm downslope position from the irrigation pipe (P_2–P_n). An additional profile was excavated at 15 cm downslope, filling the gap between the 30 cm position and the area directly beneath the pipe (P_1) (Fig. 1). To avoid edge effects, a 5 cm margin on both the left and right sides of each profile was excluded. Only the central 50 cm × 50 cm area was photographed using a digital camera (SONY Alpha 6400, Sony, Tokyo, Japan) for subsequent distortion correction and image analysis.

2.4. Dye pattern analysis

Following geometric correction, the stained and unstained regions in each soil profile image were identified, distinguished, and extracted using Photoshop CS6 (Adobe Systems Inc., San Jose, CA, USA) for further analysis (Luo et al., 2019) (Fig. 1). In these images, represent the dye-stained pathways (preferential flow), while white areas indicate unstained regions. The resolution of all profile images was set to 100 pixels/cm^2. Dye coverage (DC, %) was calculated as the dye-stained area ratio of each soil profile following Luo et al. (2019). In each dye-tracing experiment, the dye coverage of vertical preferential flow (VDC, %) was calculated from the soil profile (P_0) directly beneath the pipe to characterize the features of vertical preferential flow (Fig. 1).

For lateral dye transport, LD_{max}, representing the maximum downslope transport distance of lateral preferential flow, was defined as the distance between P_0 and P_n, where P_n represents the farthest dye-stained profile from the irrigation pipe. The dye coverage of lateral preferential flow (LDC, %) was calculated individually for each profile from P_1 to P_n as the dye-stained area ratio, whereas the lateral dye volume density (LDV) was derived from the LDC values and calculated once for each dye-tracing point. These indicators were used to characterize lateral preferential flow.

Dye coverage (DC, %) can be calculated by the dye-stained area and the total area of the soil profile (Eq. (1)) as follows:

$$DC(\%) = A_{stained}/(A_{profile})\cdot 100\% \quad (1)$$

Where $A_{stained}$ is the dye-stained area (cm^2), and the $A_{profile}$ is 2500 cm^2 in our study.

Lateral dye volume density (LDV, %) is the fraction of the lateral dye volume (LV, cm^3) to the total soil volume (SV, cm^3).

$$LV(cm^3) = \sum_{i=1}^{n-1}(A_{stained,\ i} + A_{stained,\ i+1})/2\cdot dx_i \quad (2)$$

$$SV(cm^3) = A_{profile}\cdot LD \quad (3)$$

$$LDV(\%) = LV/SV\cdot 100\% \quad (4)$$

where $A_{stained,\ i}$ (cm^2) and $A_{stained,\ i+1}$(cm^2) represent the dye-stained areas of the $P_i$and P_{i+1} soil profiles, respectively, and dx_i (cm) is the downslope distance between the P_i and P_{i+1}, of which $dx_1 = 15$ cm (between P_1 and P_2). dx_2 through dx_{n-1} are all set to 5 cm, indicating that the spacing between successive soil profiles starting from P_2 is uniform at 5 cm. LD represents the cumulative downslope distance from P_1 to P_n used for calculating LDV. The maximum of LD_{max} was 70 cm at all the sites. Therefore, to ensure a uniform comparison of LDV values across all sites, LD was set to 55 cm in the calculation. Accordingly, the number of dye-stained soil profiles (n) was fixed at 10 in this study.

2.5. Soil and root sample collection and analysis

Near each dye tracing point, one soil profile (50 cm × 50 cm) was excavated, and disturbed and undisturbed soil samples were collected to analyze soil properties and root traits in five soil layers (0–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, and 40–50 cm). In this study, undisturbed soil samples were collected with cutting rings (100 cm^3) to determine soil bulk density (BD), and Polyvinyl chloride columns (10 cm in diameter and 10 cm in height) were used to collect root samples for subsequent analysis of root traits. With respect to disturbed soil, the initial soil water content (ISWC) was determined via oven drying at 105 ℃ for 24 h, the soil organic matter content (SOM) was measured through the dichromate oxidation method, and the proportion of water-stable aggregates greater than 0.25 mm ($R_{0.25}$) was measured and calculated with the wet sieving method (Kemper and Rosenau, 1986; Mei et al., 2018; Dong et al., 2022). In root trait analysis, the roots were sieved and soaked to remove attached soil particles, scanned with an EPSON Expression 10000XL 1.0, and analyzed using WinRHIZO Pro 2009c to determine the root length density (RLD, cm/cm^3) and the root surface density (RSAD, cm^2/cm^3). The roots were then dried in an oven at 65 ℃ for 48 h and weighed to calculate the root weight density (RWD, g/cm^3) (Mei et al., 2018; Luo et al., 2023; Wang et al., 2024). Each soil profile below the pipe was investigated via the use of a 5 × 5 cm grid system and the number of roots (live and dead) was counted to calculate the live root abundance (LRA) and dead root abundance (DRA) (number/dm^2) (Luo et al., 2019). Live and dead roots in soil profiles were sorted according to root integrity, with dead roots typically exhibiting dark coloration and breaking easily (Nespoulous et al., 2019). In each soil layer, three replicates were established for each soil and root sample.

2.6. Statistical analyses

Linear mixed-effects models (LMMs) were used to examine the effects of the soil layer and site type on the soil properties, root traits and VDC. All analyses were conducted with the "lme4", "lmerTest", and "emmeans" packages. Tukey's honestly significant difference (HSD) post-hoc test was employed for multiple comparisons, and the Bonferroni method was applied to correct the P-values. All LMMs were fitted and analyzed in the R (v4.5.2).

To analyze the effects of environmental factors (soil properties and root traits) on VDC, stepwise multiple linear regression and path analysis were conducted. First, the normality of all the indicators was assessed via the Shapiro–Wilk test. Non-normal data were transformed via the use of the Johnson transformation method in Minitab 15.0 (Minitab Inc., State College, PA, USA) and the arcsine square root transformation method. The relationships between the different environmental factors and VDC were then assessed via Pearson's correlation analysis to explore potential associations. Stepwise multiple linear regression was used to derive optimal empirical models for predicting changes in VDC on the basis of soil properties and root traits, with key factors selected at a significance level of $P < 0.05$. Path analysis was subsequently conducted to examine the relationships between VDC and these key factors, as well as their interrelationships. Moreover, multicollinearity among the selected key factors was evaluated using the variance inflation factor (VIF), with all the VIF values below 5 indicating the absence of severe collinearity. The differences in the LDC, LD_{max}, and LDV values among the sites were analyzed. If the datasets were normally distributed, they were analyzed via one-way analysis of variance (ANOVA) followed by Tukey's HSD post-hoc test. For datasets that did not meet the normality assumption, the Kruskal–Wallis test was applied. In addition, Spearman's correlation analysis was used to separately assess the relationships of lateral preferential flow features (LDC, LD_{max} and LDV) with soil properties and with root traits. Above all the statistical analyses mentioned above were carried out with IBM SPSS Statistics 27 (IBM Corp., Armonk, NY, USA). All two-dimensional graphs in this study were created using Origin 2026 (OriginLab Corporation, Northampton, MA, USA), whereas three-dimensional visualization was performed using Voxler 4.6.913 (Golden Software, LLC, Golden, CO,

Table 2
soil properties, root traits and the dye coverage of vertical preferential flow (VDC) as affected by site type and soil layer among varied slope positions.

Slope position	Value	Site type	Soil layer	Site type×soil layer
Upslope	BD	ns	**	ns
	ISWC	**	**	**
	$R_{0.25}$	ns	ns	ns
	SOM	**	**	**
	LRA	**	**	**
	DRA	*	ns	ns
	RWD	ns	**	ns
	RLD	ns	**	ns
	RSAD	ns	**	ns
	VDC	**	**	ns
Midleslope	BD	ns	**	ns
	ISWC	**	**	**
	$R_{0.25}$	ns	**	**
	SOM	*	**	*
	LRA	**	**	*
	DRA	**	ns	ns
	RWD	ns	**	ns
	RLD	ns	**	*
	RSAD	ns	**	ns
	VDC	**	**	ns
Downslope	BD	ns	**	ns
	ISWC	**	*	ns
	$R_{0.25}$	ns	ns	ns
	SOM	*	**	ns
	LRA	**	**	*
	DRA	**	*	ns
	RWD	ns	**	ns
	RLD	ns	**	ns
	RSAD	ns	**	ns
	VDC	ns	**	ns

Note: BD, the bulk density (g/cm^3); ISWC, initial soil water content (%); $R_{0.25}$ (%) the proportion of water-stable aggregates greater than 0.25 mm; SOM (g/kg), soil organic matter content; LRA, live root abundance (number/dm^2); DRA, dead root abundance (number/dm^2). RWD, root weight density (mg/cm^3); RLD, root length density (cm/cm^3); RSAD, root surface area density (cm^2/cm^3). "ns", "*", "**", stand for significant differences at $P > 0.05$, $P < 0.05$, $P < 0.01$, separately.

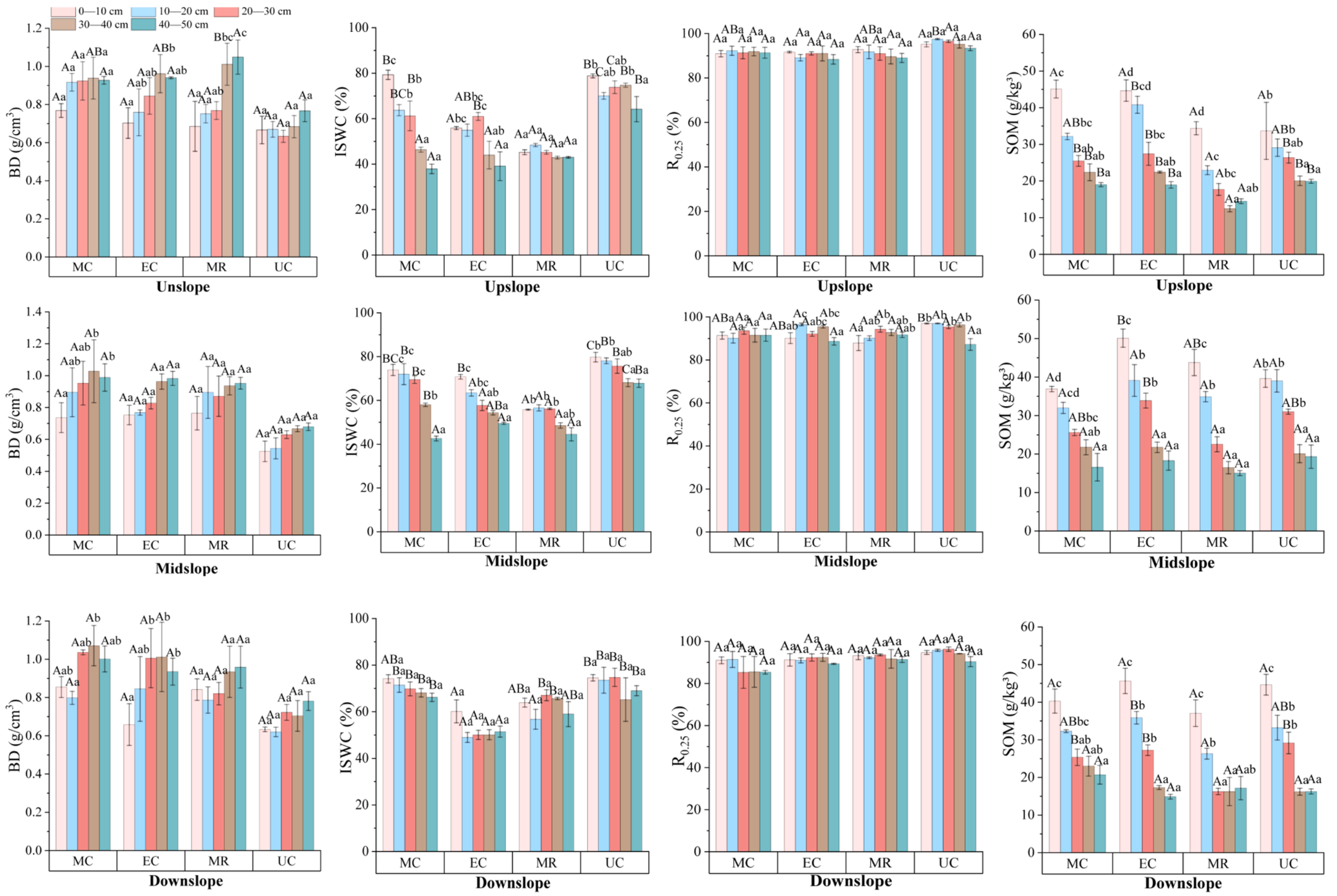

Fig. 2. Soil properties as affected by site type and soil depth. MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; UC, uncut control area; BD, the bulk density (g/cm^3); ISWC, initial soil water content (%); $R_{0.25}$ (%) the proportion of water-stable aggregates greater than 0.25 mm; SOM (g/kg), soil organic matter content; Different capital letters represent significant differences among different sites in the same soil layer ($P < 0.05$), different lowercase letters represent significant differences among different soil layers in the same site ($P < 0.05$). Each bar represents the mean of three replicates collected from the same site and soil layer. Error bars show the standard error of the mean (SE) based on the three replicates.

USA).

3. Results

3.1. Soil properties and root traits

Site type, soil layer, and their interactions exerted differential effects on soil properties and root traits (Table 2 and Figs. 2, 3 and 4). The BD, RWD, and RSAD were mainly affected by soil layer across slope positions, whereas the RLD was additionally influenced by the interaction between site type and soil layer only at the midslope position. Among these indicators, three root traits exhibited a decreasing trend with soil depth. DRA was affected only by site type at the upslope and midslope positions, whereas both site type and soil layer contributed to its variation at the downslope position, where MC and EC showed higher values than UC in the 0–30 cm layer. Meanwhile, DRA did not show uniform significant variation with soil depth across all slope positions and sites. At the upslope, SOM under the MC and EC was significantly higher than under MR in the 20–50 cm soil layer, whereas on the midslope, SOM in EC was significantly greater than in UC only in the 0–10 cm layer. ISWC was significantly higher in UC than in EC across all slope positions. Overall, LRA was higher in MR and UC than in MC and EC, and in the 0–10 cm soil layer, the difference between UC and MC was statistically significant. $R_{0.25}$ showed a depth-related response mainly at the midslope position. Overall, across the three slope positions, the SOM and LRA decreased along soil depth, whereas ISWC and $R_{0.25}$ exhibited no clear trend.

3.2. Vertical and lateral preferential flow features

Pronounced vertical and lateral preferential flow paths was observed across all site types (Figs. 5 and 6). In general, the greatest horizontal extended flow was observed in the 0–10 cm soil layer in the UC. Compared with those in the MC and EC, the quantity of preferential flow pathways in the UC and MR was low, but the width was large. Moreover, these areas were interconnected to easily form continuous patches. In terms of lateral preferential flow, compared with those in the UC and MR, the variation in lateral flow pathways within the MC and EC was low, whereas lateral migration occurred at relatively large distances. Concurrently, only a few large macropores were found to facilitate preferential flow along the lateral direction.

At the upslope and midslope positions, VDC was significantly influenced by both the site type and soil layer, whereas at the downslope position position, it was affected only by the soil layer (Table 2 and Fig. 7 (a)). At the upslope position in the 0–50 cm soil layer, the VDC in UC was significantly greater than that of both EC and MC, whereas the VDC of MR was significantly greater than that of MC. At the midslope position, UC showed significantly greater VDC than MC in the 0–20 cm and 30–50 cm soil layers. However, on the downslope, no significant differences were observed among the site types in the same soil layers. Across all slope positions, the decreasing trend in VDC with increasing soil depth was more notable in UC and MR. The LD_{max} was greater in MC

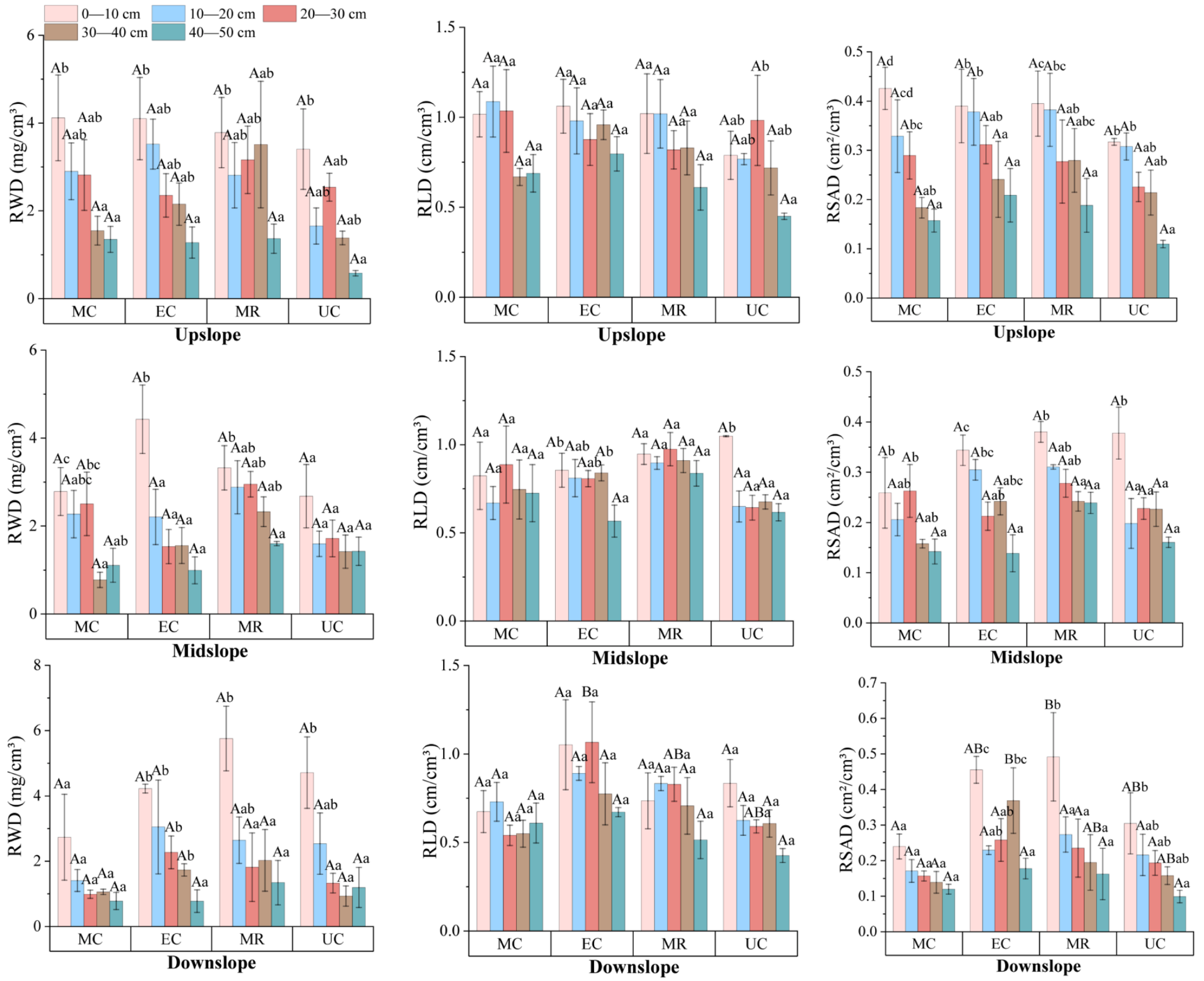

Fig. 3. Root weight density (RWD, mg/cm^3), root length density (RLD, cm/cm^3), and root surface area density (RSAD, cm^2/cm^3) as affected by site type and soil depth. MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; UC, unhcut control area; Different capital letters represent significant differences among different sites in the same soil layer ($P < 0.05$), different lowercase letters represent significant differences among different soil layers in the same site ($P < 0.05$). Each bar represents the mean of three replicates collected from the same site and soil layer. Error bars show the standard error of the mean (SE) based on the three replicates.

than in UC only at the downslope position (Fig. 7(b)).

3.3. Interactions among soil properties, root traits and preferential flow

The correlations among soil physicochemical properties, root traits, and preferential flow features are shown in Figs. 8 and 9, as well as Tables 3 and 4. Overall, VDC had significantly negative correlation with BD and DRA, but positive and correlation with all the other measured soil properties and root traits except RLD. Furthermore, indicators such as ISWC, $R_{0.25}$, SOM, and LRA were generally positively correlated with one another. ISWC, $R_{0.25}$ and LRA were extremely significant negatively correlated with DRA. Among the root traits, RWD, RLD, RSAD, and LRA also demonstrated strongly positive intercorrelations.

According to the stepwise multiple linear regression analysis results, these variables, namely, LRA, DRA and BD influenced VDC most significantly (Tables 3 and 4). Furthermore, all three variables yielded Pvalues less than 0.05, demonstrating that the regression model was statistically significant. Considering the direct effects, the LRA exerted the most notable, positive and direct influences on the VDC followed by DRA (−0.319, $P < 0.001$), and the BD (−0.227, $P < 0.001$). In terms of indirect negative effects, the BD influenced VDC most substantially by affecting LRA.

Both LD_{max} and LDV were highly significantly positively correlated with DRA and notably negatively correlated with $R_{0.25}$. Moreover, the LD_{max} was significantly negatively correlated with the LRA and positively correlated with the SOM. Furthermore, the LDC was significantly negatively correlated with the SOM.

4. Discussion

4.1. Responses of soil properties and root traits to microsites in strip-cutting areas

Strip cutting creates distinct microsites, including MC, EC, and MR sites, which can affect soil properties and root traits (Table 2, Figs. 2, 3, 4). Our study results revealed that the ISWC in the EC was significantly lower than that in UC. This occurred because an intact canopy was retained in the UC, which, through the provision of shading and

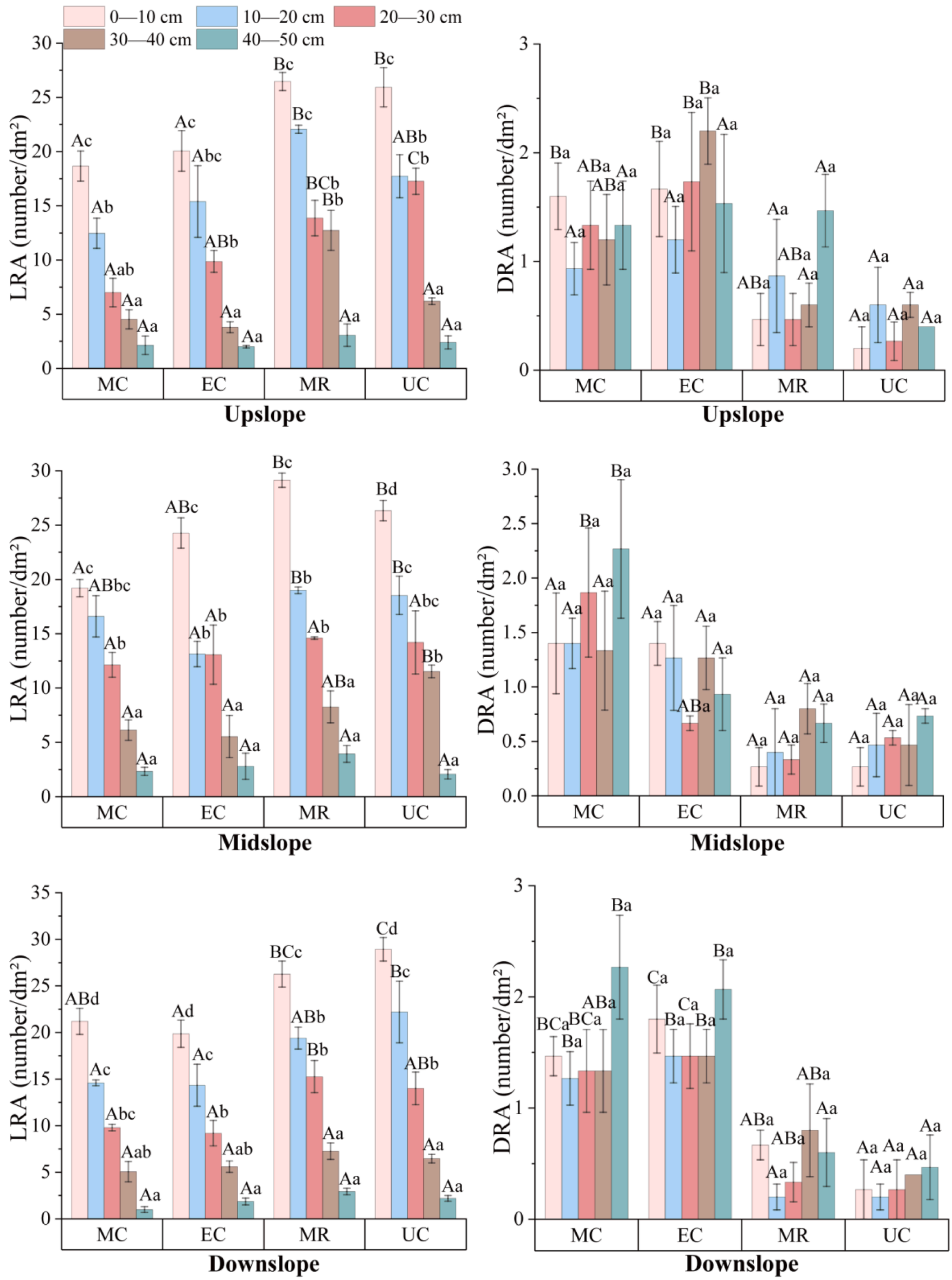

Fig. 4. Live and dead root abundance as affected by site type and soil depth. MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; UC, unhcut control area; LRA, live root abundance (number/dm^2); DRA, dead root abundance (number/dm^2). Different capital letters represent significant differences among different sites in the same soil layer ($P < 0.05$), different lowercase letters represent significant differences among different soil layers in the same site ($P < 0.05$). Each bar represents the mean of three replicates collected from the same site and soil layer. Error bars show the standard error of the mean (SE) based on the three replicates.

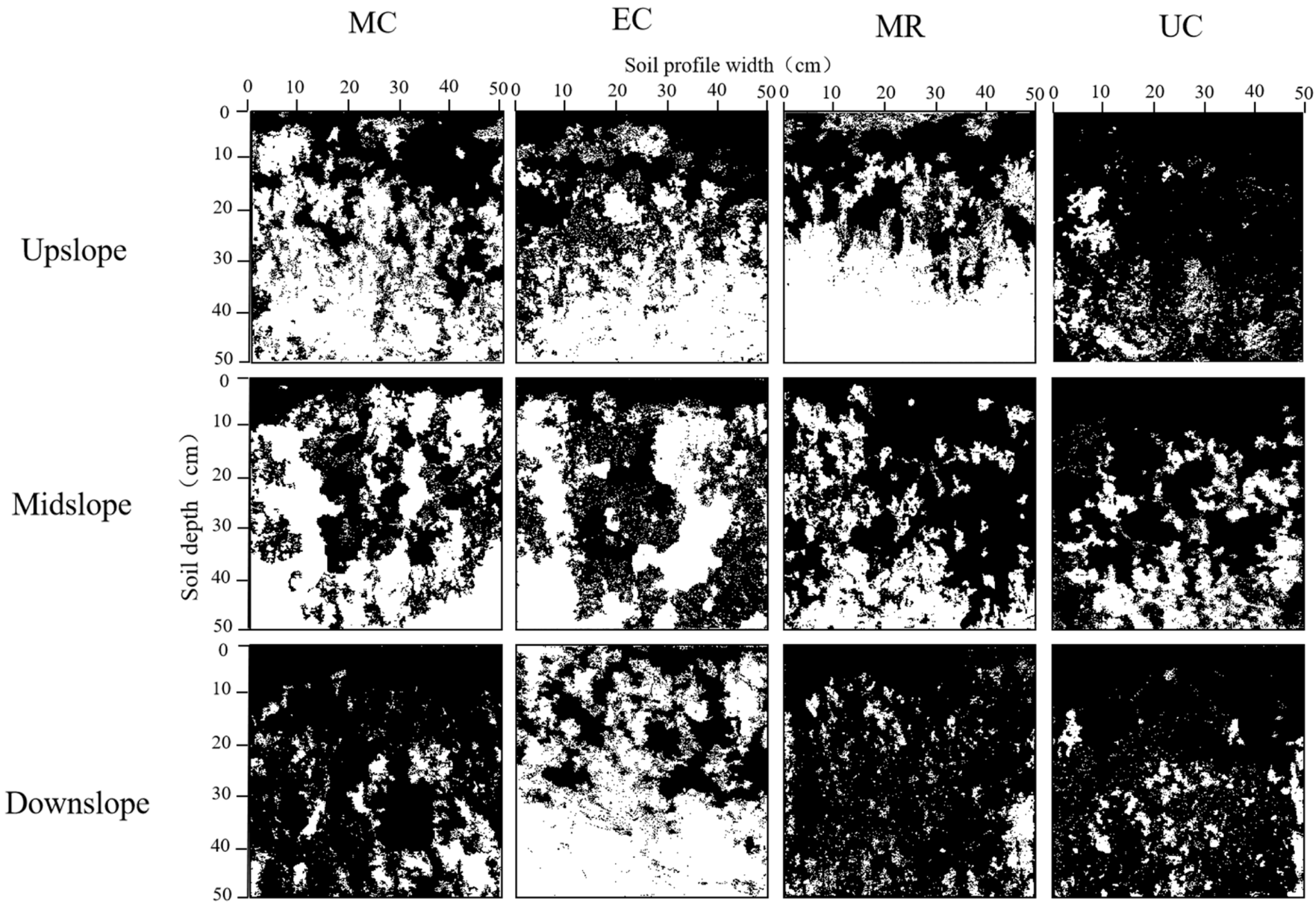

Fig. 5. Dye-stained pathways of vertical preferential flow among sites across slope positions (take replicate 1 as an example). MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; UC, uncut control area.

microclimatic buffering, reduced soil evaporation and facilitated the retention of more soil water (Meijers et al., 2025; Song et al., 2025). In contrast, the EC, as a forest-edge microsite, experienced higher soil evaporation and transpiration levels, resulting in a lower ISWC value (Herbst et al., 2007). With respect to SOM, the differences between the cutting area (MC and EC) and MR/UC varied with soil depth, a pattern that was itself slope-dependent. After strip cutting, the harvested residues retained on the soil surface decomposed and contributed to increased SOM levels (Song et al., 2025). At the upslope position, higher solar radiation, more frequent trampling, surface runoff, and erosion reduced SOM storage (Qiu et al., 2021; Xiao et al., 2024; Li et al., 2025). Although broad-leaved tree seedlings were replanted in the cutting area, the removal of aboveground vegetation and the aforementioned factors led to a greater effect than that in MR, resulting in a higher SOM only in the deep soil layers in the upslope cutting area. Simultaneously, the topographic gradient facilitates the downward transport of fine particles and organic-rich sediments via surface runoff, which typically settle in the midslope areas (Fissore et al., 2017; Zhou et al., 2024). Consequently, differences in the SOM values at the midslope positions between the EC and the UC were observed mainly in the surface soil layers (Fig. 2). At the downslope position, additional soil water recharge from the upslope and midslope positions, together with environmental variations induced by the topography (Yu et al., 2020; Ma et al., 2022; H. Liu et al., 2025), influenced the inputs and mineralization of SOM, ultimately resulting in no significant differences in the SOM among the sites. Additionally, our study results revealed found that all root traits except DRA were decreased along the soil layer, which can be attributed to root decomposition being influenced by multiple soil biotic and abiotic factors, rather than solely by the soil depth (Solly et al., 2014). Compared with MR and UC, the lower LRA in the cutting area was most evident in the surface soil layer. This may be because live roots are mainly concentrated in the surface soil (Luo et al., 2019), making them more vulnerable to harvesting-induced disturbance. DRA was significantly greater in the downslope cutting area in 0–30 cm soil layers than UC. This pattern can be attributed to the combined effects of logging disturbance and high downslope water redistribution (H. Liu et al., 2025).

4.2. Responses of vertical and lateral preferential flow to microsites in strip-cutting areas

In the microsites created by strip-cutting, the features of vertical and lateral preferential flow exhibited variation across different slopes (Table 2, Figs. 5, 6, 7). Overall, the results of this study revealed that, compared with that in the cutting area (MC and EC), the horizontal spread of the dye tracer in the surface soil of UC was greater, and the VDC in UC and MR typically decreased more obviously with increasing soil depth (Figs. 5 and 7(a)). These results are consistent with our findings that, although the number of vertical preferential flow paths in UC and MR was smaller, their widths were greater. This pattern is primarily attributable to the effects of harvesting disturbance, which reduces the quantity and continuity of macropore in cutting area (Ampoorter et al., 2007; Nazari et al., 2021). Consequently, in UC and MR, the more continuous macropores facilitated greater water and solute transport vertically through the soil layer.

At the upslope and midslope positions, the VDC of UC and MR was higher than that of EC and MC. VDC was highest in UC, significantly greater than in MC and EC, while MR showed significantly higher VDC

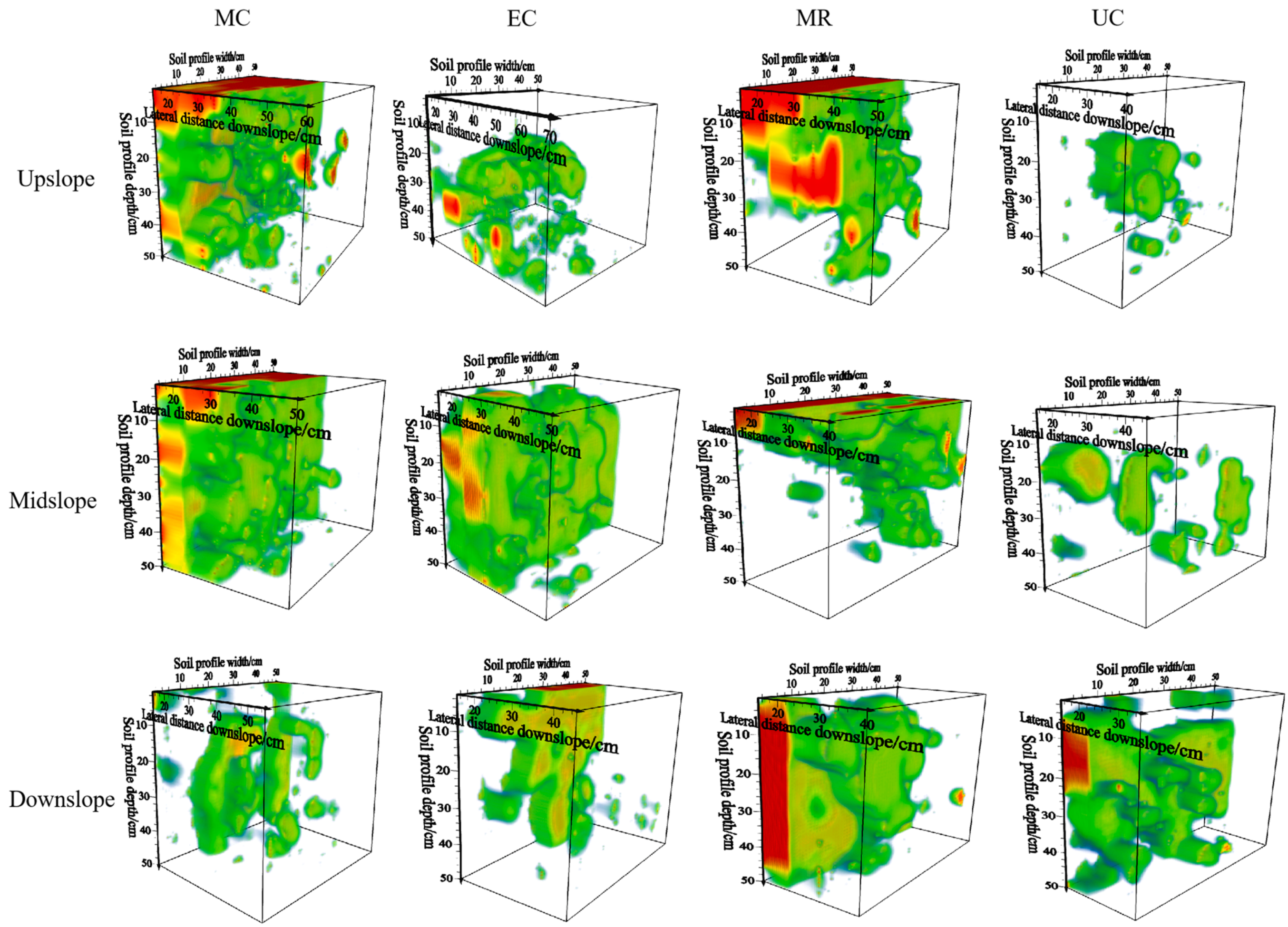

Fig. 6. Three-dimensional dye-stained pathways of lateral preferential flow among sites across slope positions (take replicate 1 as an example). MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; UC, uncut control area.

than MC at the upslope position. In most soil layers at the midslope position, the VDC of UC was significantly higher than that of MC (Fig. 7 (a)). In this study, the differences in soil properties and root traits among sites were mainly reflected in ISWC, SOM, LRA, and DRA, all of which were highly significantly correlated with VDC (Figs. 2, 4, and 8). As reported in a previous study, a higher ISWC, which is closer to the threshold water pressure, can lubricate preferential flow pathways, reduce water infiltration into the soil matrix, and facilitate the movement of water from the matrix into preferential flow channels (Kung et al., 2006; Guo and Lin, 2018; Y. Liu et al., 2025). SOM may facilitate infiltration by improving soil structure and pore connectivity (Huang et al., 2024). However, in this study, site-related differences in SOM were inconsistent across slope positions and soil layers, suggesting that its effect on VDC may be localized rather than uniformly expressed across the slope. Moreover, path analysis further showed that, compared with DRA and BD, LRA was identified as the most positive direct factor influencing VDC (Tables 3 and 4). It has been reported that live roots penetrate and disrupt the soil structure, thereby creating pores between soil particles (Qu et al., 2025). In addition, root hairs contribute to soil cementation, which increases the porosity of aggregates further. Meanwhile, root water uptake may induce shrinkage in the surrounding soil and create voids within the rhizosphere (Ghestem et al., 2011). Together, these processes promote water infiltration and conductivity (Colombi et al., 2018; Banwart et al., 2019; Lucas et al., 2019; Wang et al., 2020). Meanwhile, DRA showed a strong negative direct effect on VDC, with an effect size second only to that of LRA. This may be because fine roots on slopes tend to grow more parallel to the slope surface and generally have relatively high turnover and decomposition rates. After fine roots die, either naturally or following harvesting disturbance, the resulting dead-root channels may develop better connectivity along the slope rather than in the vertical direction (Ghestem et al., 2011; Nespoulous et al., 2019; Y. Xu et al. 2025). Consequently, these dead root channels may contribute less to the continuity of vertical soil pores. In this study, BD indirectly affected VDC by influencing LRA (Table 4), which is consistent with the findings of Nazari et al. (2021), who reported that a high bulk density can impede the root growth of plants and trees by restricting oxygen diffusion and nutrient uptake, thereby limiting root development. Although, the observed significant correlations between VDC and $R_{0.25}$, SOM, RWD and RSAD suitably agree with previous findings (Zhang et al., 2015; Wang et al., 2018; Kang et al., 2023; Cai et al., 2024; H. Liu et al., 2025; Qu et al., 2025; Wen et al., 2025; Y. Xu et al., 2025) (Fig. 8), they are not the key factors driving the differences in VDC across the various sites. This may have been due to anthropogenic disturbances in the study area, which influenced the final results. As the slope position changed from upslope to downslope, the differences in VDC between sites decreased, consistent with the patterns of soil properties and root traits among the sites along the slope. This can be attributed to two main factors. First, the thicker soils at the upslope and midslope position provide ample space for root systems and soil fauna, thus increasing the spatial variability in soil infiltration characteristics (Zhang et al., 2023; H. Liu et al., 2025; J. Xu et al., 2025). Second, at the downslope position, the low drainage capacity may alter the degree of overall infiltration (H. Liu et al., 2025) DRD.

Soil structure, SOM, and root status are the important factors of

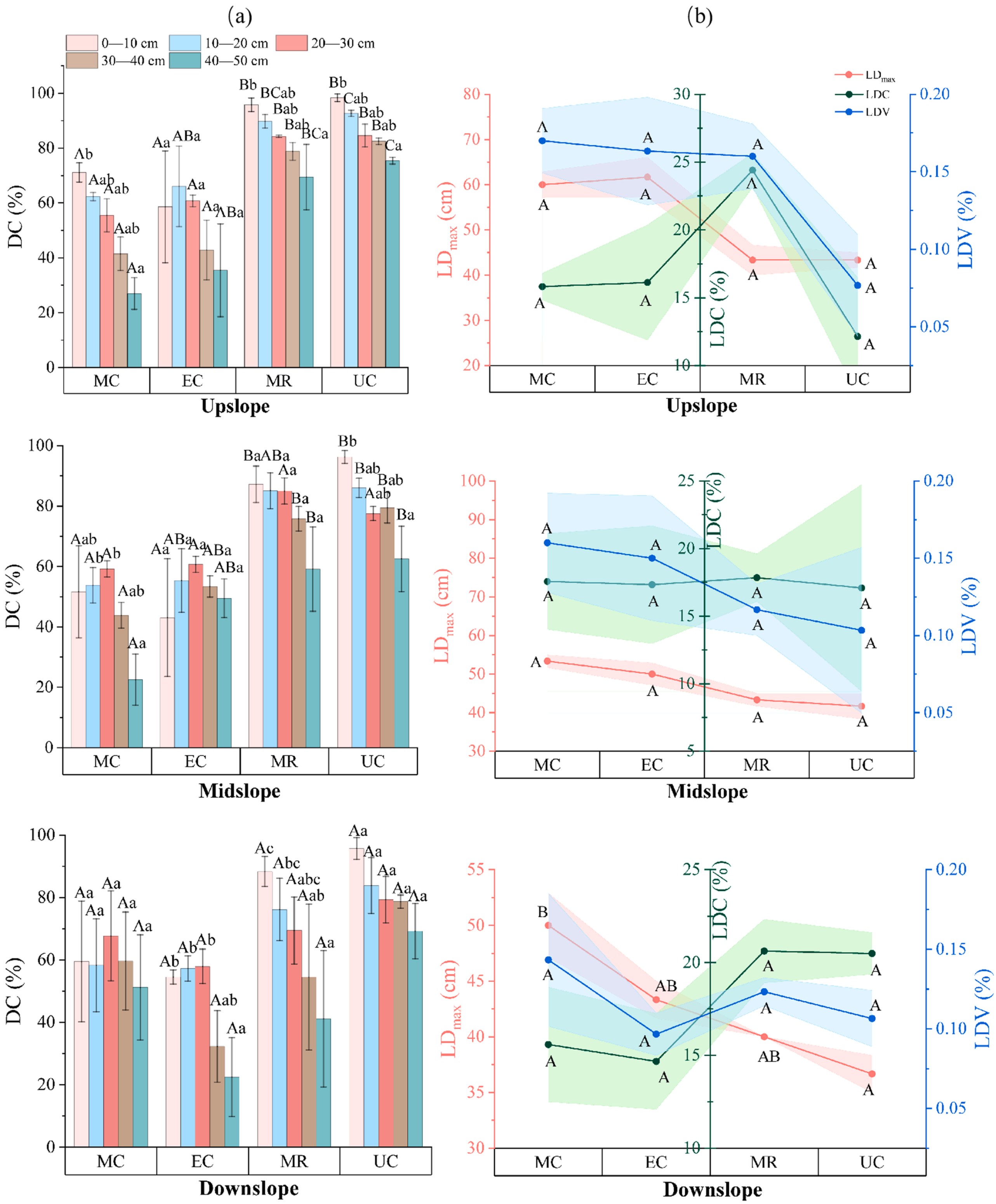

Fig. 7. Features of preferential flow among sites across slope positions. (a), vertical preferential flow features as affected by site type and soil depth. (b), lateral preferential flow features. VDC, dye coverage of vertical preferential flow (%). LD$_{max}$, maximum downslope transport distance of lateral preferential flow (cm); LDC, dye coverage of lateral preferential flow (%); LDV, lateral dye volume density (%); Different capital letters represent significant differences among different sites in the same soil layer ($P < 0.05$), different lowercase letters represent significant differences among different soil layers in the same site ($P < 0.05$). MC, middle of the cutting area; EC, edge of the cutting area; MR, middle of the retained area; UC, uncut control area. For (a), each bar represents the mean VDC calculated from three replicates within the same site and soil layer. Error bars show the standard error of the mean (SE) based on the three replicates. For panel (b), each data point represents the mean of three replicates collected from each site. Shaded areas represent the standard error of the mean (SE) for each lateral preferential flow feature (three replicates per feature).

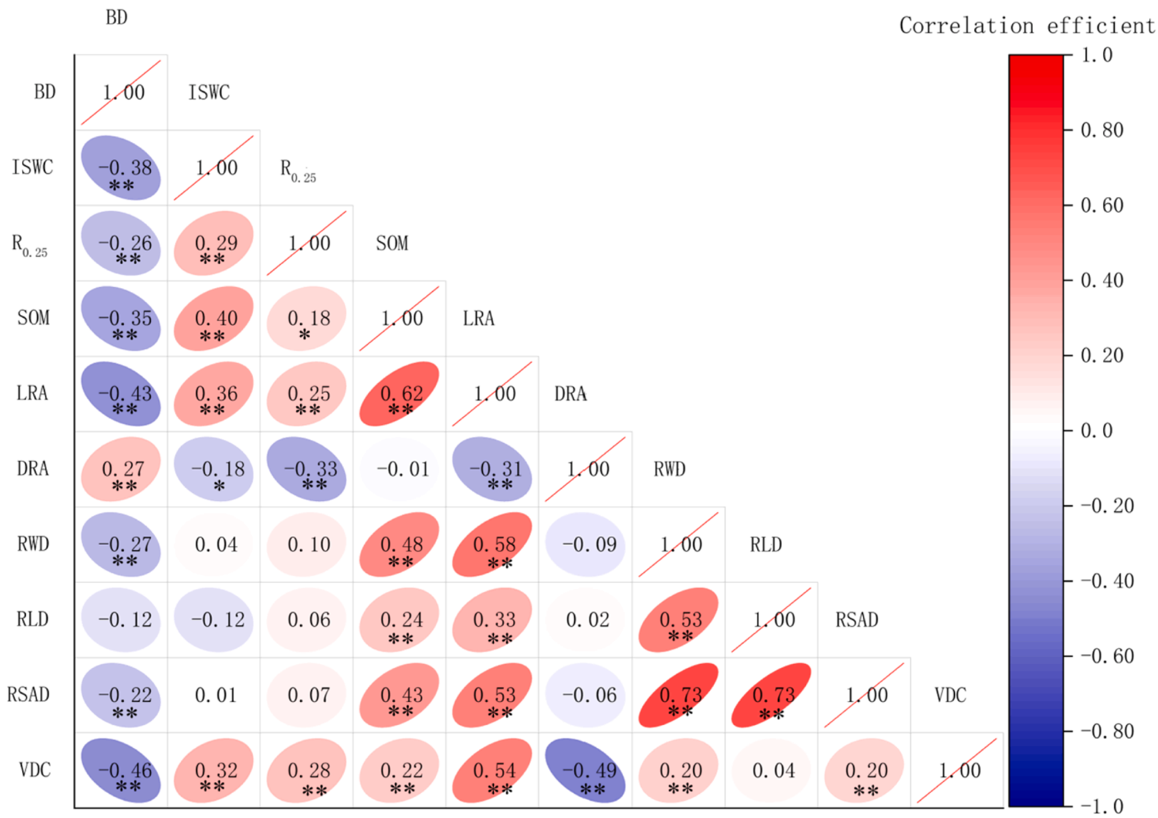

Fig. 8. Correlation analysis of VDC, soil properties and root traits. VDC, dye coverage of vertical preferential flow (%); BD, the bulk density (g/cm^3); ISWC, initial soil water content (%); $R_{0.25}$ (%), proportion of water-stable aggregates greater than 0.25 mm; SOM (g/kg), soil organic matter content; LRA, live root abundance ($number/dm^2$); DRA, dead root abundance ($number/dm^2$). RWD, root weight density (mg/cm^3); RLD, root length density (cm/cm^3); RSAD, root surface area density (cm^2/cm^3). "**", indicates significant differences at $P < 0.01$, "*", indicates significant differences at $P < 0.05$.

lateral preferential flow (Fig. 9). Soils with higher $R_{0.25}$ may indicate greater aggregate stability. Under unsaturated conditions, water can be retained within intra-aggregate pores, which may reduce the amount of water flowing into well-connected lateral macropores and thereby limit lateral preferential flow (Ferreira et al., 2023). This mechanism may partly explain why $R_{0.25}$ was significantly negatively correlated with LD_{max} and LDV (Fig. 9). Furthermore, existing evidence suggests that soil organic matter can enhance soil water retention by improving soil structure, increasing porosity, and providing additional water-adsorbing surfaces (Yang et al., 2014; Lal, 2020). This also agree with our previous findings (Fig. 8) and may partly explain the positive association between SOM and LD_{max}. Meanwhile, LDC was negatively correlated with SOM, suggesting that higher soil organic matter may be associated with fewer but longer and better-connected preferential pathways for soil water flow.

In this study, LD_{max} showed a highly significant negative correlation with LRA and a highly significant positive correlation with DRA, whereas LDV was significantly positively correlated with DRA. Compared with $R_{0.25}$ and SOM, the absolute values of the correlation coefficients between LD_{max} and LRA or DRA were larger, suggesting that live and dead roots may be more closely associated with the extension of lateral preferential flow (Fig. 9). Compared with the ring- or semi-ring-shaped gaps that form around living roots due to wetting–drying cycles, seasonal drought, or wind-induced movement, larger dead-root channels are more likely to provide open, low-resistance pathways for preferential flow, rather than promoting local diffusion and matrix infiltration (Ghestem et al., 2011; Wu et al., 2021; Qu et al., 2025). Meanwhile, clusters of living fine roots may absorb and temporarily retain soil water. After root death and decomposition, these structures may contribute to high pore-water pressure nodes, especially along dead-end flow paths, with potential implications for slope stability (Ghestem et al., 2011; Benegas et al., 2014; Zhang et al., 2020). In general, lateral roots that grow parallel to the hillslope create more continuous and permeable pathways near the soil surface, which facilitates the development of lateral preferential flow. This, in turn, can lead to enhanced solute transport and potentially contribute to slope instability (Noviandi et al., 2025). Overall, these findings may help explain why elevated DRA in the 0–30 cm soil layer of downslope MC was associated with greater LD_{max}.

4.3. Contributions, limitations, and future research

By using an in situ dye-tracing experiment, this study simultaneously visualised vertical and lateral preferential flow across four sites (three strip-cutting microsites and uncut control site) and slope positions, and further visualised the three-dimensional stained pathways of lateral preferential flow. This approach enriched the quantitative indicators for evaluating this lateral preferential flow. Our results revealed that, compared with other soil properties and root traits, live roots were a key factor promoting vertical preferential flow, whereas dead roots were significantly positively related to the maximum downslope transport distance of lateral preferential flow. These findings provide new insight into the mechanisms through which root status regulates preferential flow pathways on forested hillslopes affected by harvesting disturbance. Previous studies have indicated that lateral flow is key to slope stability and containment transport (Buttle and McDonald, 2002; Leslie and Heinse, 2013; Bourgault et al., 2017; Nespoulous et al., 2019; Y. Xu et al., 2025). Therefore, following harvesting disturbance, forest management should not only focus on soil fertility and stand growth, but also on long-term monitoring of how temporal changes in root status affect vertical and lateral preferential flow, as these two flow pathways may further influence soil erosion, water redistribution, and nutrient

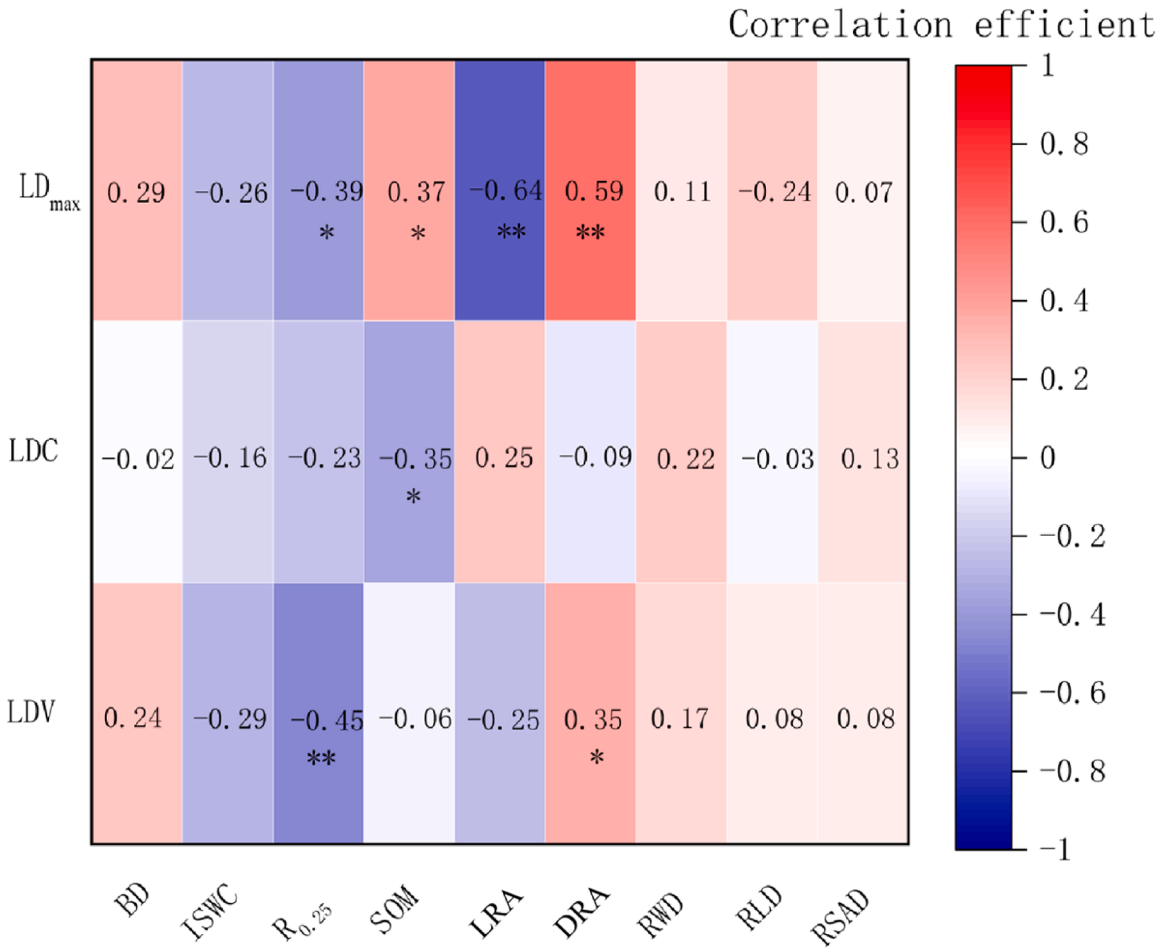

Fig. 9. Correlation analysis of lateral preferential flow features with soil properties and root traits. LD_{max}, maximum downslope transport distance of the dye tracer (cm); LDC, dye coverage of lateral preferential flow; LDV, lateral dye volume density (%); BD, bulk density (g/cm^3); ISWC, initial soil water content (%); $R_{0.25}$ (%), proportion of water-stable aggregates greater than 0.25 mm; SOM (g/kg), soil organic matter content; LRA, live root abundance (number/dm^2); DRA, dead root abundance (number/dm^2). RWD, root weight density (mg/cm^3); RLD, root length density (cm/cm^3); RSAD, root surface area density (cm^2/cm^3). "**", indicates significant differences at $P < 0.01$, "*", indicates significant differences at $P < 0.05$.

transport.

In this study, due to methodological limitations, the flow velocities of vertical and lateral preferential flow could not be measured simultaneously with high temporal precision. Moreover, the classification of live and dead roots relied primarily on observations of soil profiles, and the three-dimensional anisotropy of root channels across different root types and different diameter classes remains poorly characterised. Future studies should combine multiple scales and employ non-destructive, high-precision methods to accurately distinguish live and dead roots in situ and quantify their spatial distributions across diameter classes, in order to better assess their respective contributions to vertical and lateral preferential flow. Apart from these issues, future studies should also integrate hillslope hydrological survey methods, such as ERT, soil moisture monitoring, and hydrometric tracer techniques, with

Table 3
The stepwise multiple linear correlation between change in VDC and its corresponding variance.

Dependent variable	Linear equation	Adjisted R^2 (%)	P-value
VDC	y = 0.368 + 0.340LRA - 0.319DRA - 0.227BD	0.432	< 0.001

Note: VDC, dye coverage of vertical preferential flow (%); BD, bulk density (g/cm^3); LRA, live root abundance (number/dm^2); DRA, dead root abundance (number/dm^2).

Table 4
Path analysis of the direct and indirect effects of soil properties and root traits on VDC.

Variable	Direct path coefficient	LRA	DRA	BD	Indirect path coefficient	Sum of coefficient	P-value
LRA	0.340		0.100	0.097	0.197	0.537	< 0.001
DRA	−0.319	−0.106		−0.061	−0.167	−0.486	< 0.001
BD	−0.227	−0.145	−0.086		−0.231	−0.458	<0.001

Note: VDC, dye coverage of vertical preferential flow (%); BD, bulk density (g/cm^3); LRA, live root abundance (number/dm^2); DRA, dead root abundance (number/dm^2).

well-established root study approaches. Under the context of synchronized long-term rainfall monitoring, these methods would allow a more dynamic assessment of the relationships between root traits and preferential flow.

5. Conclusion

This study provides insights into vertical and lateral preferential flow in hillslope soils of *C. japonica* plantations (23-years-old) within strip-cutting areas (30 m cutting + 45 m retained, 40%, intensity), as well as the interactions between soil properties and root traits. At the upslope, midslope, and downslope positions, microsites (MC, EC and MR) created by strip-cutting exhibited depth-dependent variations in soil and root characteristics. At the upslope position, SOM in MC and EC was significantly greater than that in MR at 20–50 cm, whereas on the midslope, SOM in EC exceeded that in UC in the 0–10 cm layer. The ISWC was consistently higher in UC than in EC across all slope positions. Differences in root traits among the sites were mainly observed in terms of LRA and DRA. Notably, LRA in UC at 0–10 cm was greater than in MC, whereas DRA in MC and EC at downslope position was higher than UC at 0–30 cm. VDC was greater in UC than MC across most soil layers at the upslope and midslope position, whereas LD_{max} in MC exceeded that in UC at downslope. LRA and DRA showed opposite relationships with VDC and LD_{max}. Specifically, LRA had a highly significant positive effect on VDC but was negatively correlated with LD_{max}, whereas DRA showed the opposite pattern. These findings indicate that live and dead roots play contrasting roles in regulating vertical and lateral preferential flow, with live roots promoting vertical preferential flow but limiting lateral transport distance, whereas dead roots show the opposite effect. This highlights the importance of monitoring root status and its effects on preferential flow components after strip-cutting. Compared with that in UC, the extended lateral preferential flow in downslope MC may have substantial implications for solute transport and slope stability, warranting long-term monitoring. These findings provide a mechanistic understanding of preferential flow pathways and offer a theoretical basis for ecohydrological research and slope stability assessment in managed plantation forests on hilly landscapes.

CRediT authorship contribution statement

Ziteng Luo: Writing – review & editing , Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shenshen Zhang:** Visualization, Data curation. **Chunxiu Liu:** Visualization. **Yufeng Liu:** Visualization. **Shuqin He:** Conceptualization. **Yunqi Zhang:** Conceptualization. **Kuangji Zhao:** Funding acquisition, Data curation. **Haiyan Yi:** Data curation. **Yong Wang:** Writing – review & editing, Funding acquisition, Conceptualization. **Bo Tan:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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